[1] NTT Basic Research Laboratories

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QCrypt 2015@Tokyo

All-photonic quantum repeaters

Koji Azuma[1]

Collaborators:
Kiyoshi Tamaki[1]
Hoi-Kwong Lo[2]

Innovative R&D by NTT
Quantum Cybernetics
FIRST 最先端研究開発支援プログラム
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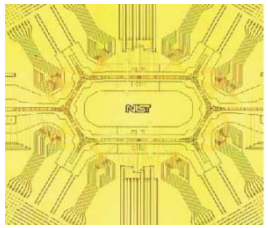
KA, K. Tamaki & H.-K. Lo
Nature Commun. **6**, 6787 (2015).

Realization of quantum information processing

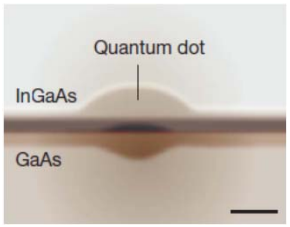
Quantum computing

Pic from Nature **464**, 45 (2010)

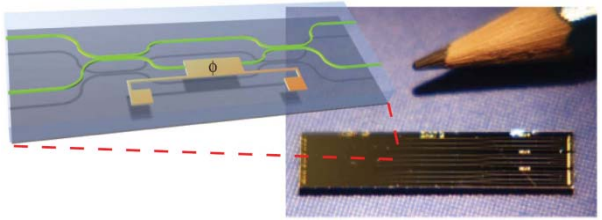
Computation is NOT fastidious about the types of physical system.



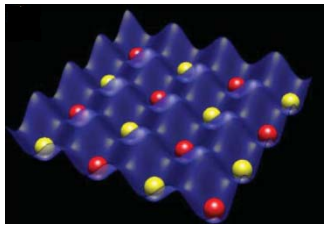
Ion trap



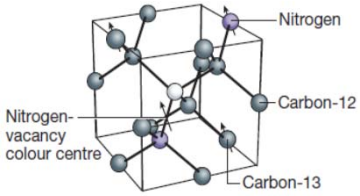
Quantum dot



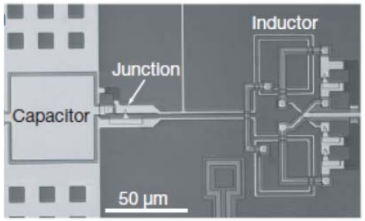
Photon



Atom in optical lattice



NV center in diamond

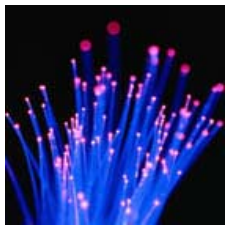


Super-conducting qubit

Quantum communication

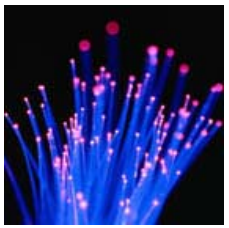
But, communication is fastidious. Photons are almost only candidates.

Direct transmission ~ 300 km Quantum repeaters → distance



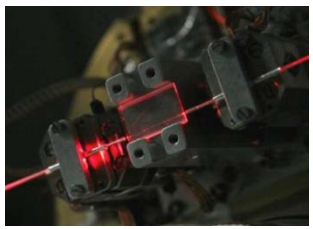
Photon via fiber

(Pic from Phys.org [Dec. 05, 2012])



Photon via fiber

\+



Matter quantum memories

(Pic from Scientific American [Jan.13, 2011])

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Realization of quantum information processing

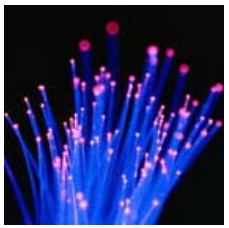
Only quantum repeaters enforced us to use systems in a hybrid manner.

- Photons
- Matter quantum memories
  - Ensemble system
    - Infinite coherence time is required. Razavi *et al*., PRA **80**, 032301 (2009)
    - How to make it fault tolerant (although it is not a qubit system)?
  - Qubit system
    - It is required to satisfy not only Divincenzo's five criteria but also his extra.
      D. P. Divincenzo, quant-ph/0002077
    - Is this much more demanding than quantum computing?

Quantum communication

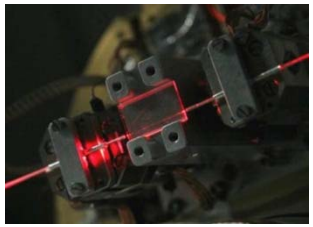
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Quantum repeaters may be more difficult than a quantum computer, if we follow the dogma of the necessity of matter quantum memories.

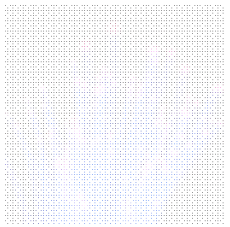
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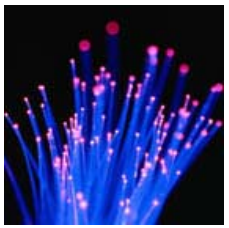
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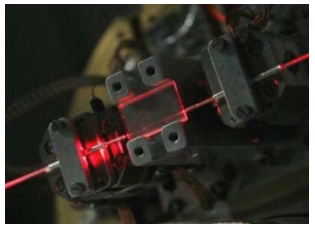
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Matter quantum memories

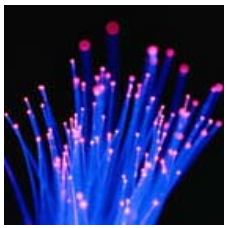
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Innovative R&D by NTT

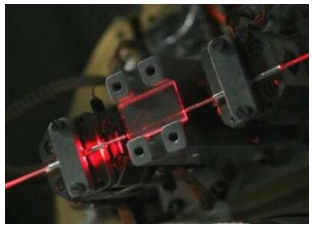
Quantum repeaters may be more difficult than a quantum computer,
if we follow the dogma of the necessity of matter quantum memories.

However, today, we show a new quantum repeater scheme, which is much simpler than a quantum computer and is the ultimate of all optical network.

Quantum repeaters ⟶ distance



\+



Photon via fiber Matter quantum memories

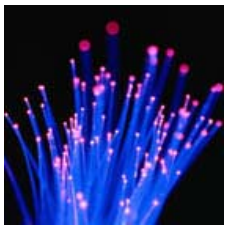
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Innovative R&D by NTT

All-photonic quantum repeaters

KA, K. Tamaki & H.-K. Lo, Nat. Commun. **6**, 6787 (2015).

Quantum repeaters → distance



Photon via fiber

+

Innovative R&D by NTT

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Nature Commun. **6**, 6787 (2015).

All-photonic quantum repeaters

KA *et al*., Nat. Commun. **6**, 6787 (2015).

This protocol uses only

- Linear optical elements
- Single-photon sources
- Photon detectors
- Fast active feedforward techniques

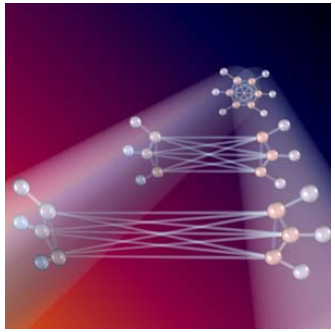


Image from the web site of Nature Commun. [http://www.nature.com/ncomms/archive/date/2015/04/index.html]

Distinguished advantages:

- Repetition rate of this protocol could be increased as high as one wants.
- Coherent frequency converters for photons could be unnecessary.
- All the elemental components are simpler than matter quantum memories.
- It could work at room temperature.
- It is proved to be much simpler than the KLM quantum computer.

The main idea:

"Time reversal" of DLCZ-type quantum repeaters.

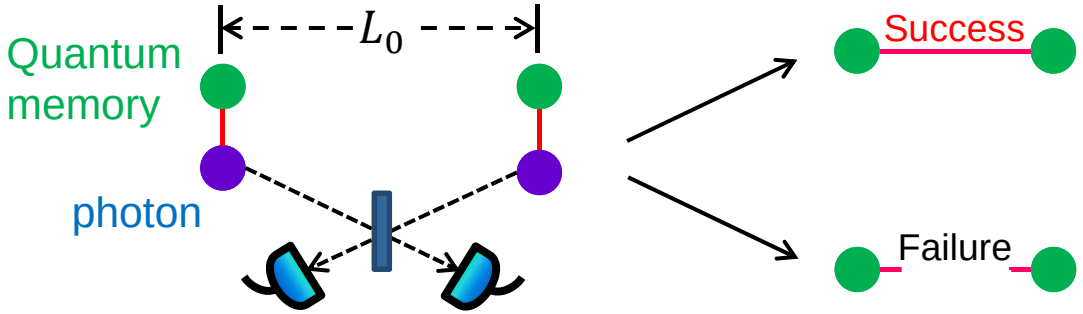
Innovative R&D by NTT

KA, K. Tamaki & H.-K. Lo
Nature Commun. **6**, 6787 (2015).

Review: DLCZ quantum repeaters

L.-M. Duan *et al.*, Nature **414**, 413 (2001).

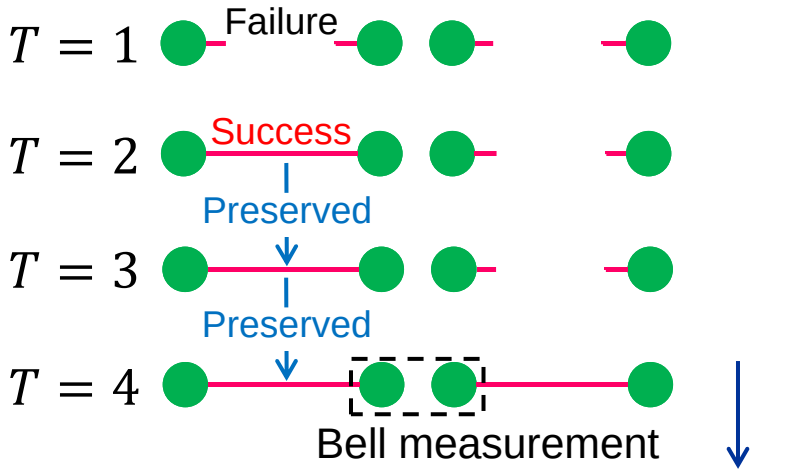
(1) Single-photon-based entanglement generation



(2) Entanglement swapping

Original DLCZ-type proposals:

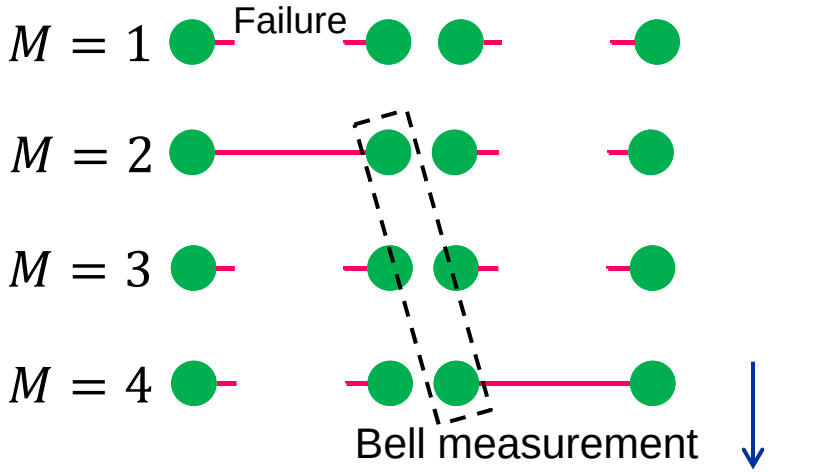
For ideal quantum memory,



With $M = 1$

Multiplexed version:

To relax the requisite for memory time.



At $T = 1$

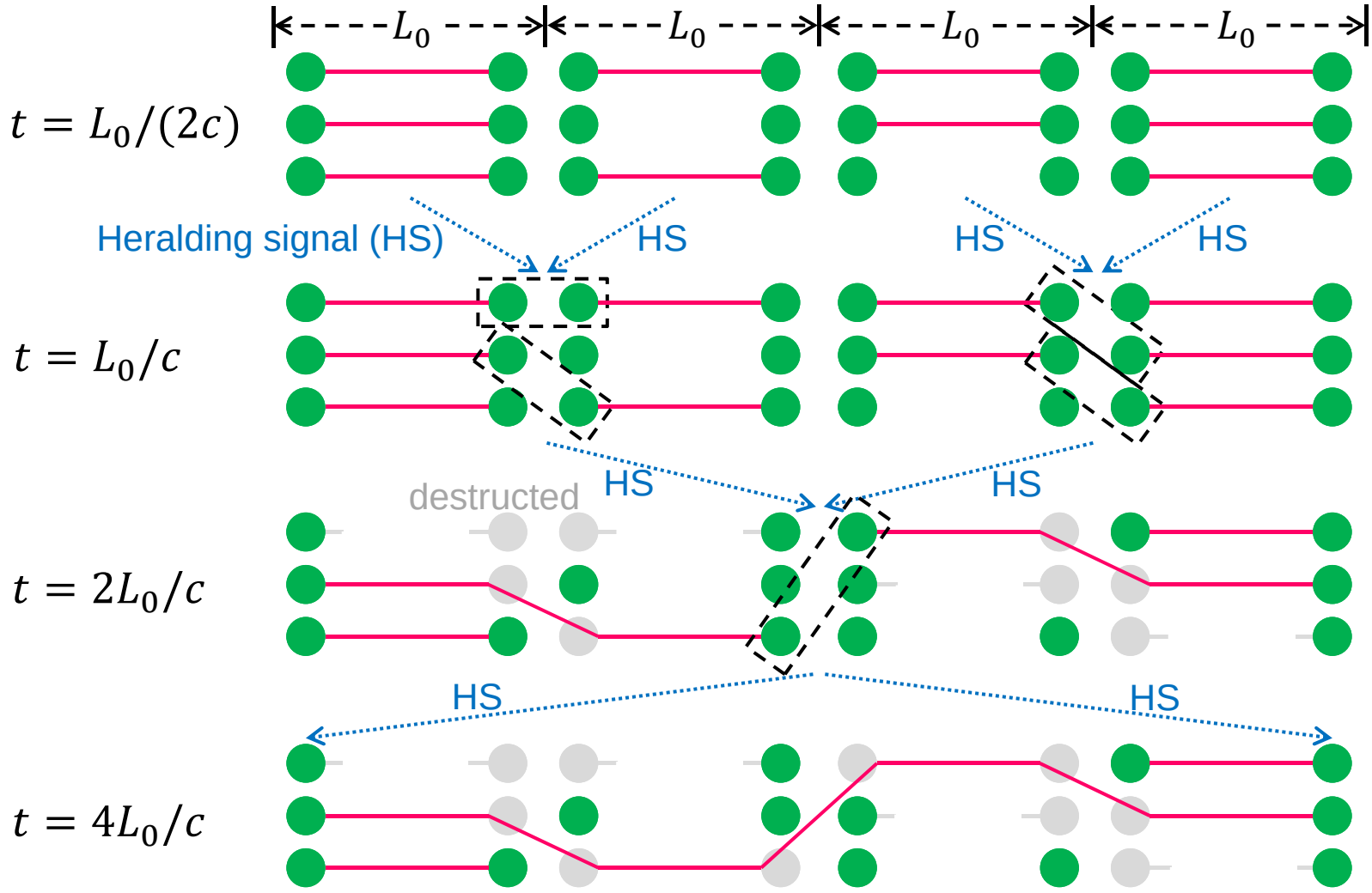
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DLCZ-type quantum repeaters with realistic quantum memories

Even if we use the multiplexing, we still need quantum memories. Why?

➡ Heralding signals (HSs) are needed to be exchanged over long distances.



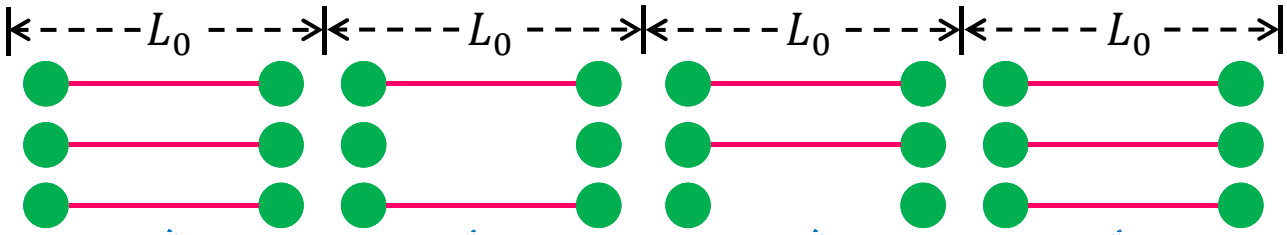
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$t = L_0/(2c)$

Notice

1) Entanglement swapping succeeds **only probabilistically.**

$t \longrightarrow$ **Waiting time** for the HSs is proportional to **the overall distance**.

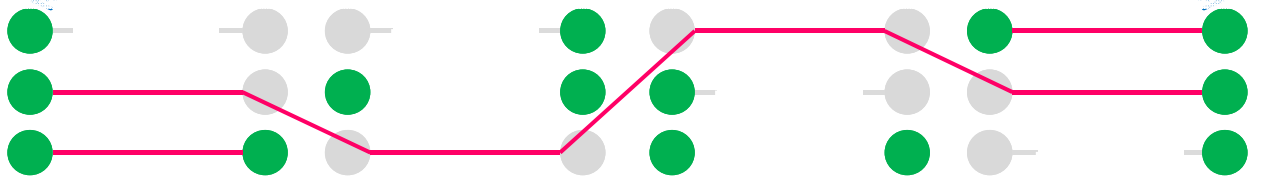
2) Dephasing or depolarization for quantum memories increases **exponentially with time**.

Exponential scaling. DLCZ-type protocol does not work, necessitating to go back to Briegel *et al.*'s protocol that needs a quantum computer.

M. Razavi *et al.*, Phys. Rev. A **80**, 032301 (2009)

time

$t = 4L_0/c$



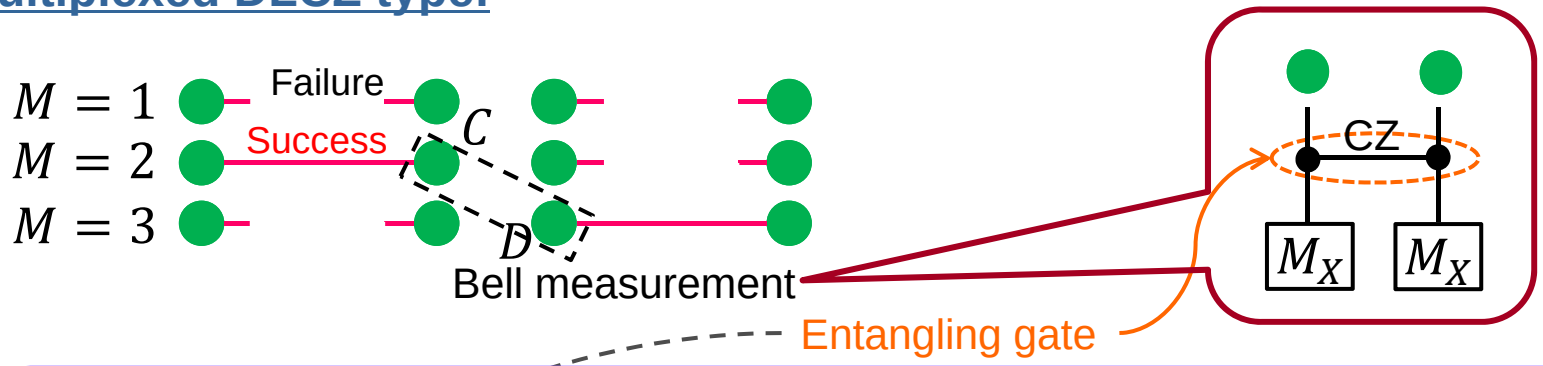
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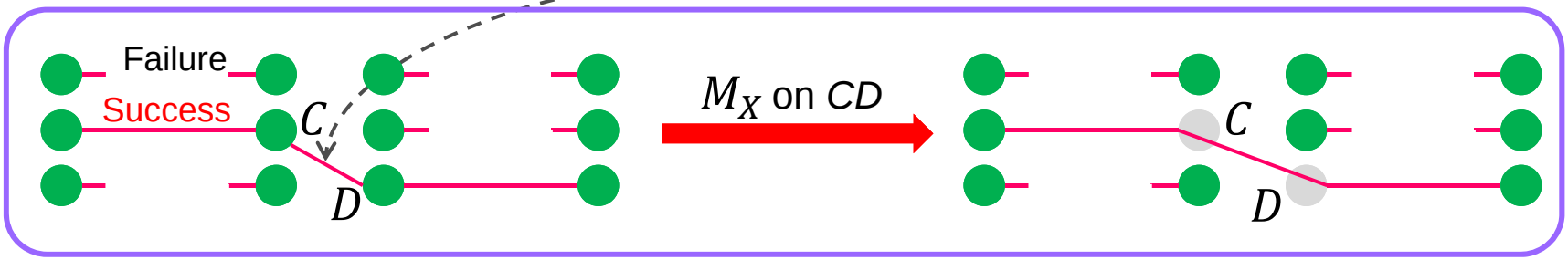
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Time reversal of DLCZ scheme

KA *et al.*, Nat. Commun. **6**, 6787 (2015).

Multiplexed DLCZ type:

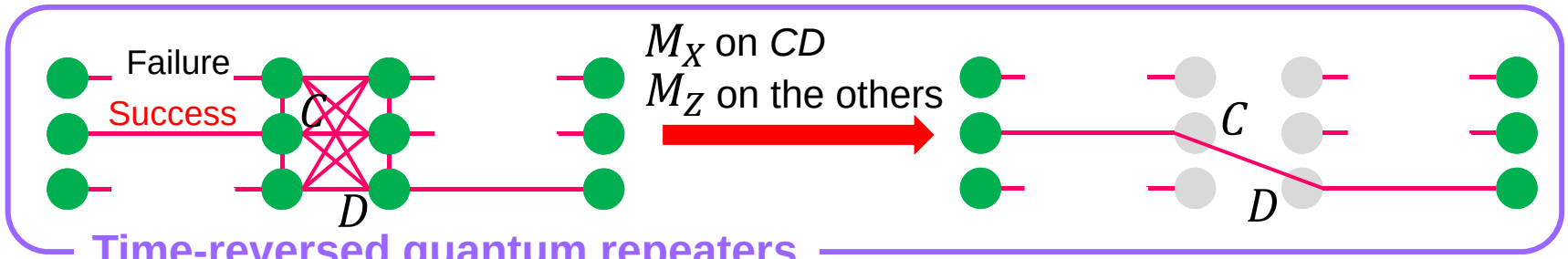




If we have entangled *C* and *D* from the first, only single-qubit measurements are needed.

deterministic in contrast to entangling gates

But we cannot initially predict which pair becomes a successfully entangled pair *CD*.
Thus, we set the qubits in a complete-like cluster state.

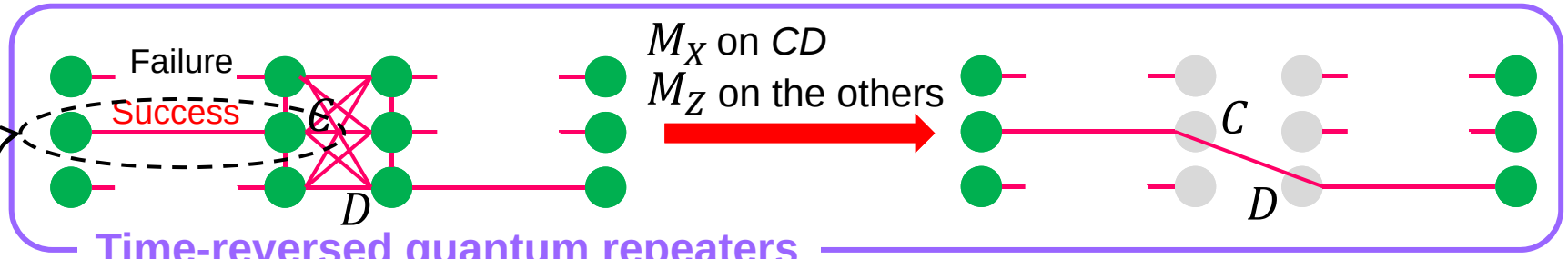


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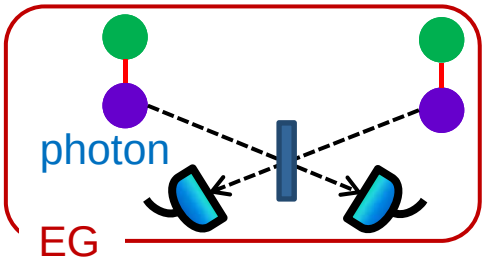
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All-photonic quantum repeaters

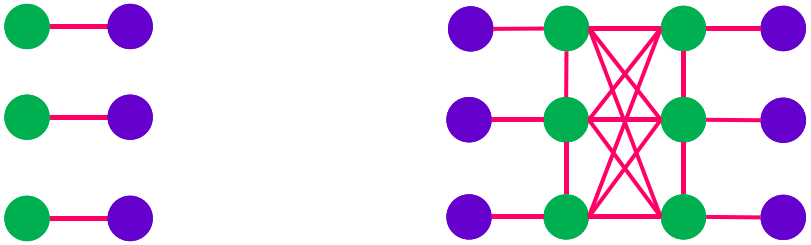
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How to supply them with entanglement? — Remember



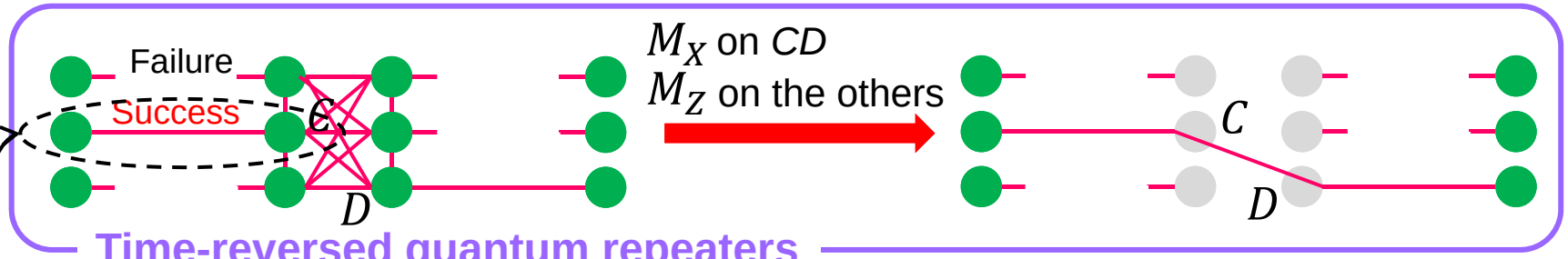
The entire picture:



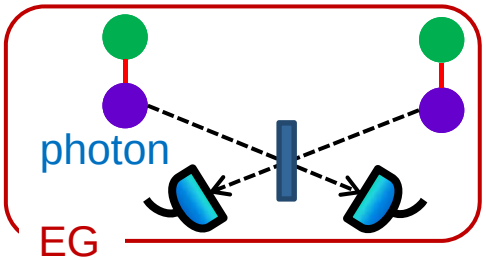
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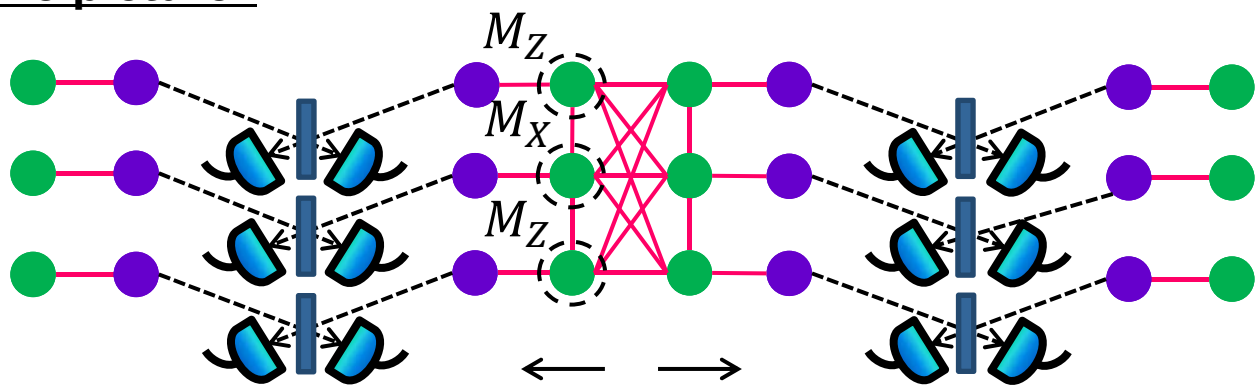
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How to supply them with entanglement? — Remember



The entire picture:



We just need M_X or M_Z on ● depending on the outcomes of Bell measurements on ●.

→We can regard ● as **single photons** that are sent with ● to know the Bell measurement outcomes.

All photonic!

Loss in the transmission

→ This is overcome by Varnava *et al.*'s code.

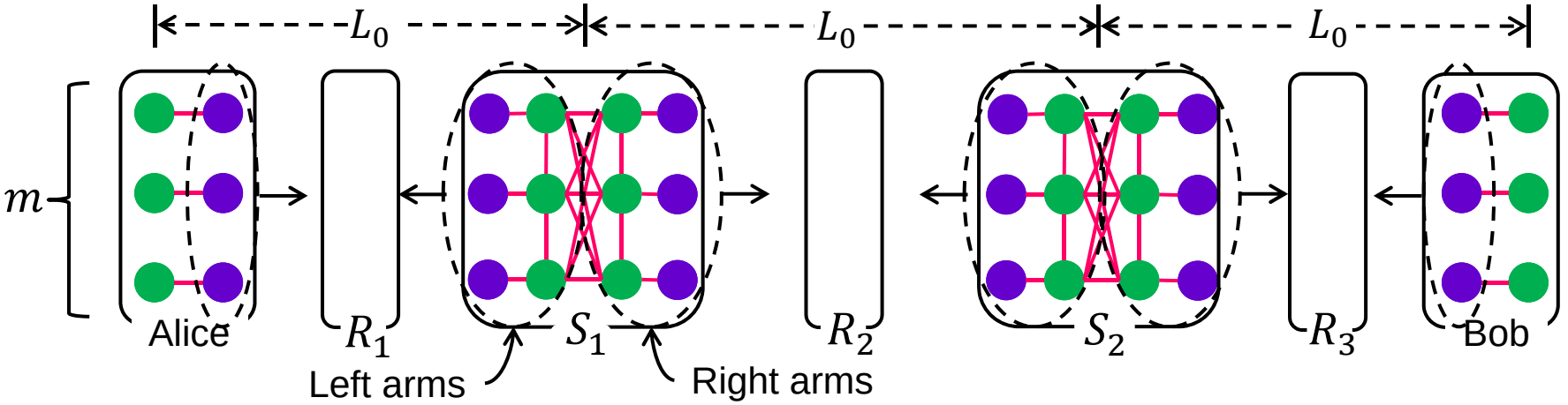
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All-photonic quantum repeaters

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1) Off-line entanglement swapping (Local preperation)



S_i: Sending node

R_i: Receiver node

●: Encoded qubit (Varnava *et al.*'s code)

●: Row qubit

- Completely-like cluster state is prepared locally with 20 time adaptive measurements with photon detectors, linear optics elements and high speed active feedforward.
  ~150 ns [Nature **437**, 116 (2005)]

  Preparation time is translated to the corresponding photon loss probability.

- The number m of arms are chosen to minimize the total photon number.

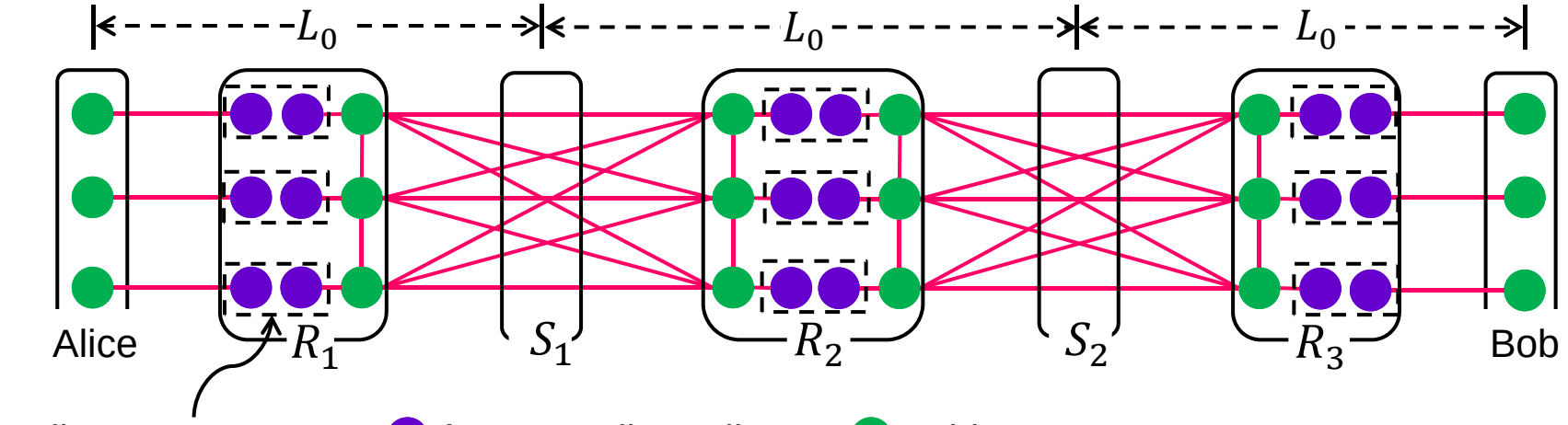
- Left (Right) arms are transmitted to the adjacent left (right) receiver nodes.

Innovative R&D by NTT

All-photonic quantum repeaters

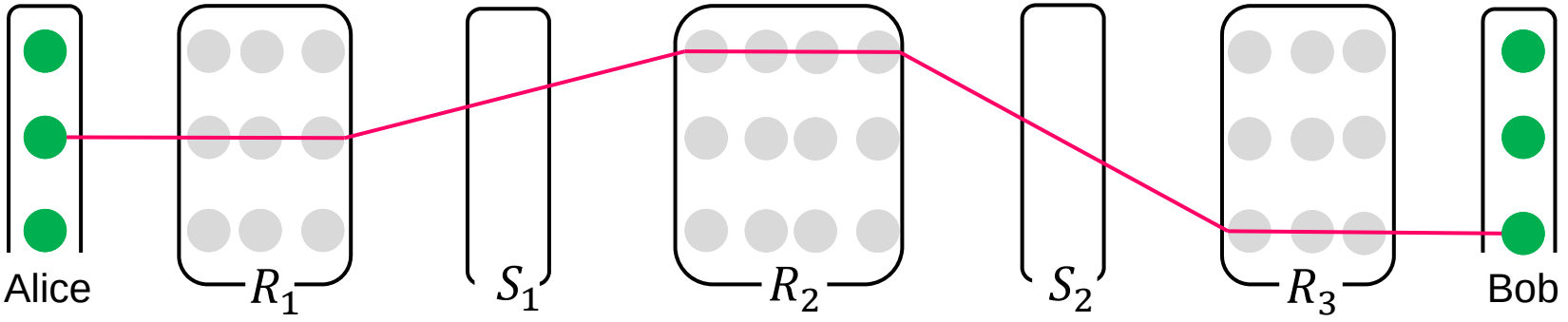
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2) Entanglement generation



Bell measurement on ● for entangling adjacent ● qubits.

Every receiver node performs
- M_X on ● of one pair of successfully entangled pairs.
- M_Z on the other pairs of qubits ●.



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The properties

KA *et al.*, Nat. Commun. **6**, 6787 (2015).

- **Polynomial scaling:**
  From the time-reversed construction of our protocol itself, the average total photon number $\bar{Q}$ consumed in the protocol to produce an entangled pair follows

$$\bar{Q} \sim \mathrm{poly}(L)$$

  for the total communication distance L.

- **Highest bandwidth:**
  The average rate $\bar{R}$ of our protocol to produce an entangled pair with a single repeater system is

$$\bar{R} \sim f$$

  for the repetition rate f of the slowest device among single-photon sources, single -photon detectors, and active feedforward techniques.

- **Robustness against transmission errors:**
  Thanks to the special robustness of Varnava *et al.*'s protocol for M_X and M_Z and to the faithful transmission of a single photon over a short distance, the error is suppressed exponentially with distance L.

Innovative R&D by NTT

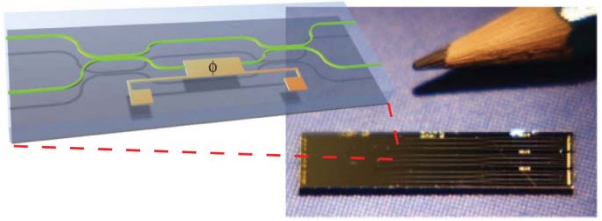
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Realization of quantum information processing

Quantum computing

Nature **464**, 45 (2010)

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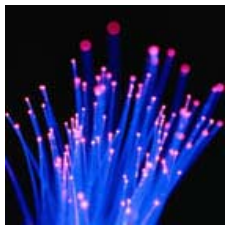
Photon

Everything is possible by photons alone.

Quantum communication

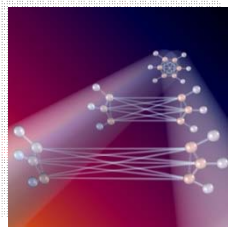
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(Pic from the web site of Toshiba Cambridge)



All-photonic quantum repeaters

(Image from the web site of Nature Commun.)

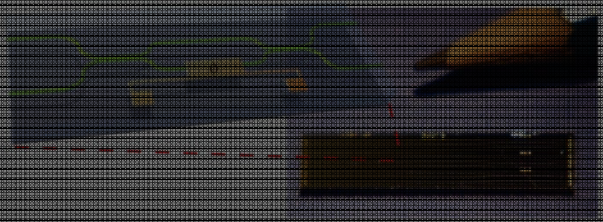
Innovative R&D by NTT

KA, K. Tamaki & H.-K. Lo
Nature Commun. 6, 6787 (2015).

Summary: Realization of quantum information processing

Quantum computing Nature **464**, 45 (2010)

Computation is NOT fastidious about the types of physical system.



Photon

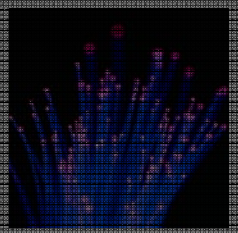
Everything should be done with photons alone.

Think with photons!

Quantum communication

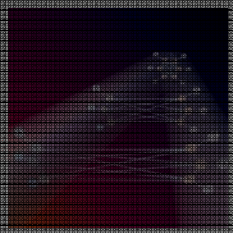
But, communication is fastidious. Photons are almost only candidates.

Direct transmission ~ 300 km Quantum repeaters → distance



Photon via fiber

(Pic from the web site of Toshiba Cambridge)



All-photonic quantum repeaters

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